

● RAG Learning

Required Practicals — OCR B GCSE Physics

OCR GCSE Physics B (J259) — Apparatus, techniques, methods, analysis. Questions on *any* practical assessed across both papers.

The OCR GCSE specification requires schools to cover a core set of practical activities. Exam papers include questions on these — you'll need to describe methods, identify independent/dependent/control variables, evaluate precision/accuracy/repeatability, and analyse results. OCR groups the required apparatus and techniques into eight **Practical Activity Groups (PAG1–PAG8)**, and you must complete at least eight practical activities that cover them all. This page follows the PAG order, using the example activity for each group plus the other practicals the specification names under that PAG. Your school may use different activities for the same PAG.

PAG1 — Materials: Density of Solids and Liquids (P6.1)

Aim: Measure the density of a regular solid, an irregular solid and a liquid.

Method: Regular solid (e.g. cuboid):

1. Measure mass with a balance (g).
2. Use a ruler or vernier callipers to measure length, width, height (cm).
3. Volume = $l \times w \times h$. Density = m / V .

Irregular solid (displacement can):

1. Fill an eureka can with water to the spout.
2. Place a measuring cylinder below the spout.
3. Gently lower the object in; catch displaced water. Volume = volume collected.

Liquid:

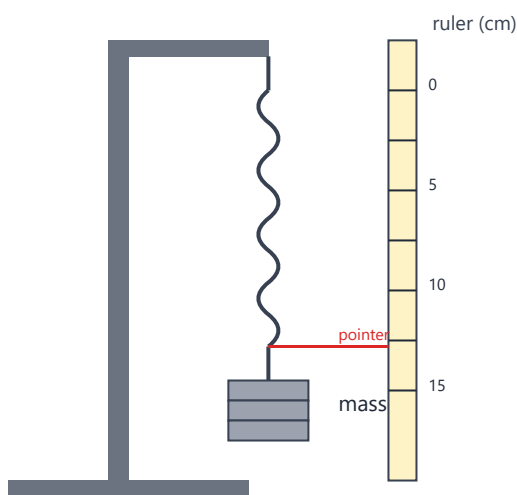
1. Place an empty measuring cylinder on the balance and zero it.
2. Pour liquid in and read volume and mass simultaneously. Density = m / V .

Exam Tip: Measure volumes in cm^3 and masses in g, then density is in g/cm^3 . Multiply by 1000 for kg/m^3 .

PAG2 — Forces: Effect of Forces on a Spring (P6.3)

Aim: Investigate the relationship between force applied to a spring and its extension.

Hooke's Law Apparatus



Method:

1. Hang a spring from a clamp stand. Attach a pointer to the bottom of the spring.
2. Measure the unloaded position l_0 with the ruler next to the pointer.
3. Add a 100 g mass ($\approx 1 \text{ N}$) to the hanger. Record the new pointer position l .
4. Add masses one at a time up to at least 6 (6 N). After each addition, record position.
5. Extension $x = l - l_0$ for each force.
6. Plot a graph of force (y) against extension (x). A straight line through origin confirms Hooke's Law. Gradient = spring constant k .

Safety: Take care — if the spring breaks it can recoil. Place a "crash pad" under the hanger in case masses fall.

PAG3 — Motion: Speed and Acceleration of a Trolley down a Ramp (P4.2)

Aim: Measure the speed of a trolley rolling down a ramp, then investigate its acceleration and how force and mass affect it.

Method: Speed down a ramp (ruler and stopwatch or light gate):

1. Prop one end of a ramp on a block. Mark a start line and measure distances along the ramp with a metre rule.
2. Release the trolley from rest at the start line (do not push it) and time how long it takes to travel a measured distance.
3. Average speed = distance $\div$ time. Repeat three times and take a mean.
4. For the speed at a point, fix a card of known length to the trolley and use a light gate:
speed = card length $\div$ time the gate is blocked.
5. Change the ramp height and repeat — a steeper ramp gives a greater speed at the bottom.

Exam Tip: Stopwatch timing includes human reaction time (about 0.2 s), which matters most for short times. Light gates remove this, and longer distances reduce the percentage uncertainty.

Method: Acceleration (light gates and a computer / data logger):

1. Set up a ramp with a trolley. Attach an interrupt card of known length to the trolley.
2. Place two light gates a fixed distance apart along the trolley's path.
3. Use a pulley system with a known weight pulling the trolley (this is the accelerating force).
4. Release the trolley. The light gates and timer record the time at each gate, giving speed before (u) and after (v) and the time between (t).

5. Calculate $a = (v - u) / t$.
6. **Vary the force** by moving masses from the trolley to the hanger (keeping total mass the same — total m constant).
7. **Vary the mass** by adding masses to the trolley while keeping the accelerating force constant.
8. Plot a against F (expect straight line through origin, confirming $F = ma$) and a against $1/m$ (expect straight line).

Safety: Keep the ramp and trolley on the bench; place a stop or cushioned box at the bottom so the trolley and falling masses cannot drop onto feet.

PAG4 — Measuring Waves: Ripple Tank and Speed of Sound (P1.3)

Aim: Use a ripple tank to measure the frequency, wavelength and speed of water waves, and measure the speed of sound in air.

Method: Ripple tank (water waves):

1. A motor with an eccentric wheel creates a continuous wave pattern on the water surface. A stroboscope or slow-motion camera freezes the pattern.
2. Measure the distance across several wavelengths and divide — reduces uncertainty. Record λ .
3. Count the number of waves passing a point in 10 s and divide by 10 $\rightarrow f$.
4. $v = f \times \lambda$.

Speed of sound in air (two microphones or echo):

1. Two microphones connected to a fast timer (or a dual-trace oscilloscope / data logger) are placed a measured distance apart, in line with the sound source.
2. Make a sharp sound (e.g. clap or strike two blocks) beyond the first microphone. The timer records the time for the sound to travel between them.

3. Speed = distance between microphones $\div$ time. Repeat and take a mean.
4. Echo method: stand about 50 m or more from a large wall, clap, and time 20 or more echoes by clapping in rhythm with them. Speed = $(2 \times \text{distance} \times \text{number of echoes}) \div \text{total time}$. Expect about 330–340 m/s.

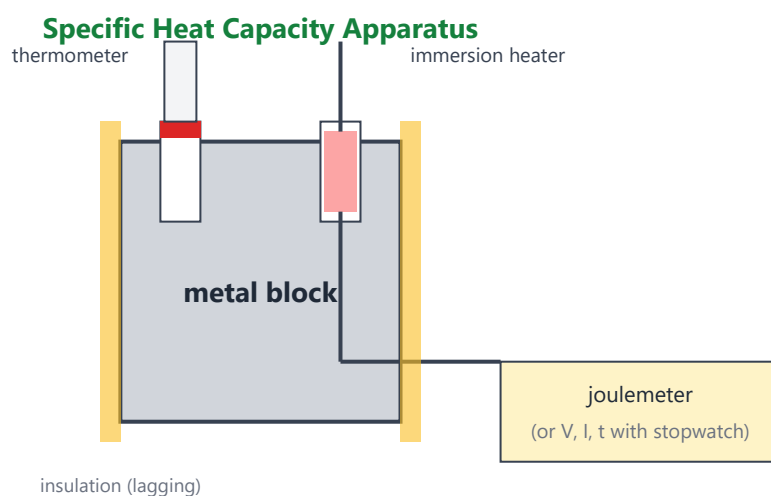
Exam Tip: A lamp above the ripple tank projects the wave pattern onto paper below, which makes the waves easier to count and measure. Measure across ten wavelengths and divide by ten to reduce uncertainty in λ .

Safety: Water near electrical equipment: keep the lamp and motor supply away from spills and mop up water immediately.

PAG5 — Energy: Specific Heat Capacity of a Metal (P6.1)

Aim: Determine the specific heat capacity of a metal block.

- **Independent variable:** time / energy supplied.
- **Dependent variable:** temperature of the block.
- **Controls:** mass of block, starting temperature, amount of insulation.



Method:

1. Measure the mass **m** of the block using a top-pan balance.
2. Insert the immersion heater and thermometer into the holes. Add a drop of oil to the thermometer hole for good thermal contact.
3. Record the initial temperature **θ_1** .
4. Connect the heater to the joulemeter (or in series with an ammeter and voltmeter). Switch on and start a stopwatch.
5. Every minute, record temperature for 10 minutes.
6. Record final temperature **θ_2** and energy input **E** from the joulemeter (or calculate $E = V \times I \times t$).

$$c = E / (m \times \Delta\theta)$$

Safety: Wear eye protection. The heater and block become hot — do not touch. Let cool before removing.

PAG6 — Circuit Components: I–V Characteristics (P3.2)

Aim: Investigate the I–V characteristics of a fixed resistor, a filament lamp and a diode, and how the resistance of an LDR and a thermistor changes.

Method:

1. Set up a circuit with the component in series with a variable resistor, ammeter and cell. Voltmeter in **parallel** with the component.
2. Adjust the variable resistor to give a range of p.d. values across the component. Record V and I at each step.
3. Reverse the cell (swap connections to battery) to obtain negative values.
4. Plot I (y) against V (x). Expected shapes:
 - Fixed resistor: straight line through origin (ohmic).

- Filament lamp: S-shape through origin (R rises with temperature).
- Diode: zero below ~ 0.7 V, then sharp rise. No current in reverse.

Method: LDR and thermistor:

1. Connect the LDR or thermistor in series with an ammeter and a low-voltage supply, with a voltmeter across it (or use an ohmmeter).
2. LDR: move a lamp to different distances (or use a light meter) and calculate $R = V / I$ at each light level. Resistance **falls** as light intensity increases.
3. Thermistor: place it in a beaker of water with a thermometer, heat in steps, stir, and calculate R at each temperature. Resistance of an NTC thermistor **falls** as temperature increases.

Exam Tip: Draw the circuit with the ammeter in series and the voltmeter in parallel with the component. A straight I–V line through the origin means constant resistance (linear); a curve means the resistance changes (non-linear).

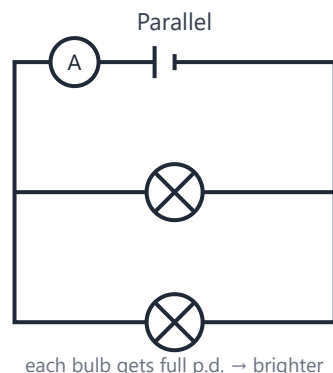
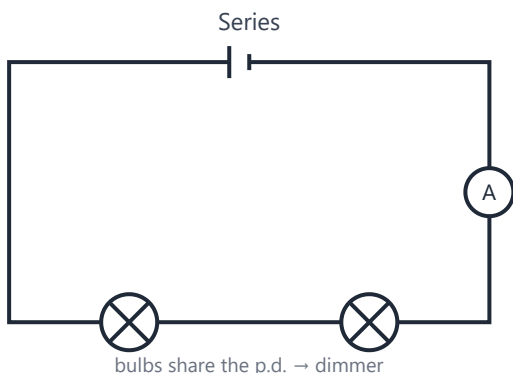
Safety: Filament lamps get hot. Use low currents so components do not overheat; take care with hot water when testing the thermistor.

PAG7 — Series and Parallel Circuits: Brightness of Bulbs (P3.3)

Aim: Compare the brightness of identical bulbs in series and in parallel, and measure the current and potential difference in each arrangement.

- **Independent variable:** circuit arrangement (series or parallel) and number of bulbs.
- **Dependent variables:** bulb brightness, current, potential difference.
- **Controls:** same supply p.d., identical bulbs.

Bulbs in Series and in Parallel



Method:

1. Build a circuit with a low-voltage supply (or battery), a switch, an ammeter and one bulb. Note its brightness and record the current and the p.d. across the bulb.
2. Add a second identical bulb **in series**. Compare the brightness, record the current (it is the same at every point in the loop) and the p.d. across each bulb.
3. Rebuild with the two bulbs **in parallel**, each on its own branch. Record the current in each branch and from the supply, and the p.d. across each bulb.
4. Optionally add a third bulb to each arrangement and repeat.
5. Brightness can be compared by eye against a reference bulb, or measured with a light meter / light sensor at a fixed distance.

Expected results: In series the supply p.d. is shared between the bulbs and the total resistance is larger, so the current is smaller and each bulb is dimmer. In parallel each bulb has the full supply p.d. across it and is as bright as a single bulb; the supply current is the sum of the branch currents. If one bulb in series breaks, all go out; in parallel the others stay lit.

Exam Tip: Series: current is the same everywhere and p.d.s add up to the supply p.d. Parallel: p.d. is the same across each branch and branch currents add up to the total current. Always draw the ammeter in series and the voltmeter in parallel.

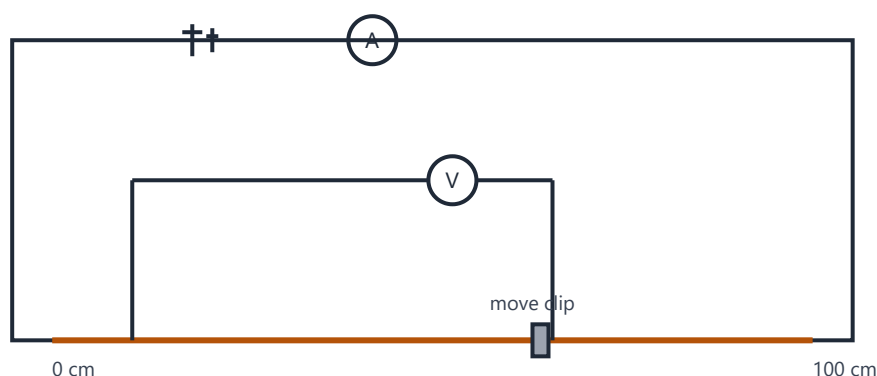
Safety: Use low-voltage supplies only. Bulbs and wires can get hot — switch off between readings and do not exceed the bulb's rated voltage.

PAG7 — Series and Parallel Circuits: Resistance of a Wire (P3.2)

The specification links this practical to PAG7, alongside the bulbs investigation above.

Aim: Investigate how the resistance of a wire depends on its length.

Resistance of a Wire — Circuit



Method:

1. Set up the circuit with a battery, ammeter in series, and a 1 m length of constantan (or nichrome) wire taped to a metre rule.
2. Connect a crocodile clip at 10 cm. Place the voltmeter across the length of wire between the fixed end and the crocodile clip.
3. Close the switch briefly, record V and I , and open the switch again (to avoid the wire heating up).
4. Calculate $R = V / I$ for this length.
5. Repeat for at least 6 different lengths (20, 30, ..., 100 cm).
6. Plot R against L . A straight line through the origin confirms $R \propto L$.

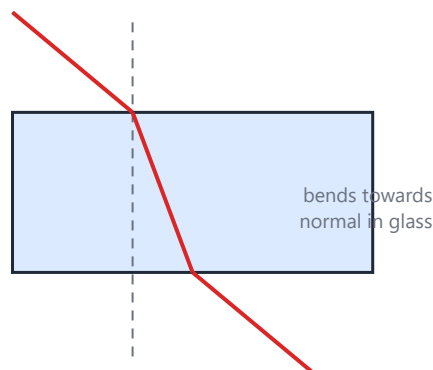
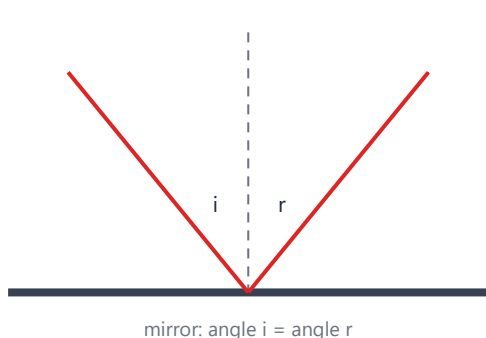
Exam Tip: Keep the current low ($< 0.5 \text{ A}$) and turn off between readings — otherwise the wire heats up and resistance rises, giving systematic error.

PAG8 — Interactions of Waves: Reflection and Refraction of Light (P1.3)

Aim: Investigate the reflection of light by a plane mirror and the refraction of light through a glass or Perspex prism (or block).

- **Independent variable:** angle of incidence.
- **Dependent variable:** angle of reflection / angle of refraction.
- **Controls:** same mirror / prism, same ray box and slit, angles measured from the normal.

Plane Mirror and Rectangular Block



Method: Reflection by a plane mirror:

1. Draw a straight line on paper and stand the plane mirror along it. Draw a normal (at 90°) to the line.
2. In a darkened room, shine a narrow ray from a ray box at the mirror where the normal meets it.
3. Mark the incident and reflected rays with small crosses, remove the mirror, and join the crosses with a ruler.
4. Measure the angles of incidence and reflection from the normal with a protractor. Repeat for several angles.

Refraction through a prism or block:

1. Place the prism (or rectangular block) on paper and draw round it. Draw a normal where the ray will enter.
2. Shine a ray at the surface; mark the incident ray and the ray leaving the other face.

3. Remove the block, join the entry and exit points to show the ray inside, and measure the angle of incidence and angle of refraction from the normal.
4. Repeat for a range of angles of incidence. With a triangular prism and white light, also observe the spectrum produced by dispersion.

Expected results: For a plane mirror the angle of reflection equals the angle of incidence. When light enters glass it slows down and bends **towards** the normal; on leaving it speeds up and bends **away** from the normal. A ray along the normal does not change direction.

Exam Tip: Measure all angles from the **normal**, not from the surface. Using a narrow ray and sharp pencil reduces uncertainty; marking the ray at two points far apart makes its direction more accurate.

Safety: Ray box lamps get hot — do not touch the bulb housing. Handle the mirror and glass blocks carefully in case of chipped edges, and work in dim light with the room otherwise clear.

Other Practical Work (not a PAG)

The following activity is suggested in the radioactive materials chapter but is not one of the eight Practical Activity Groups.

Half-life Analogue with Dice (P5.1)

Aim: Model radioactive decay using dice or coins.

Method:

1. Start with 100 dice in a beaker. Each die = one "atom".
2. Roll all the dice onto the table.
3. Remove the dice that show a 6 — these have "decayed".
4. Count the remaining dice, record, and repeat.

5. Plot number remaining (y) against throws (x). A smooth exponential curve. Use the graph to read off how many throws it takes for the count to fall to 50, 25, 12.5 — these intervals are equal, confirming constant half-life.

This analogy illustrates that decay is **random** but **statistically predictable** — exactly the pattern observed in real radioactive samples.

Practical Skills — Everything You Need to Know

Skill	What examiners expect
Variables	Identify independent, dependent and control variables
Accuracy	How close to the true value (use calibrated instruments, correct zero errors)
Precision	Spread of repeated readings (small spread = precise)
Repeatability	Same person, same equipment → similar results
Reproducibility	Different person/equipment → similar results
Resolution	Smallest change a measuring instrument can detect
Anomalies	Readings that don't fit the pattern — investigate and possibly exclude from the mean
Uncertainty	$\pm$ half the resolution for analogue; $\pm$ the last digit for digital (rule of thumb)

Exam Tip: Common mark-scheme hooks: "measure with a *suitable* instrument" (e.g. micrometer for wire diameter), "repeat and take a mean", "plot a graph to identify a relationship", "identify a control variable", "evaluate the limitations of the method".